

Scenario 1 – Feature Selection

Question:

A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has 50 features, but the team wants to use only the most relevant ones. How can they apply feature selection effectively?

Concept: Feature Selection

Step-by-Step Logic:

1. **Collect the Data** – Collect diabetes patient data such as **Age, Glucose, Blood Pressure, BMI, Insulin** and the **Outcome**.
2. **Separate X and y** – Use patient details as **Features (X)** and diabetes outcome as **Target (y)**.
3. **Apply Feature Selection** – Use **SelectKBest with Mutual Information** to measure feature importance.
4. **Select Top Features** – Rank the 50 features and select the most relevant features, such as the **top 10**.
5. **Train and Predict** – Train the model using the selected features and predict diabetes for new patients.

Scenario 2 – Django Project

Question:

You're creating a new project for managing student data. The project name is `student_portal`. What is the step-by-step command to start it?

Concept: Django Project Creation

Step-by-Step Logic:

1. **Open Terminal** – Open Command Prompt or Terminal.
2. **Check Django** – Make sure Django is installed in the environment.
3. **Go to Project Location** – Navigate to the folder where the project should be created.
4. **Create Project** – Use the Django command to create the `student_portal` project.
5. **Enter the Project Folder** – Move into the `student_portal` directory.
6. **Start Server** – Start the Django development server and open it in the browser.

Scenario 3 – Dimensionality Reduction

Question:

A company is analyzing customer purchase patterns with 200+ behavioral features. How can they reduce dimensionality without losing predictive power?

Concept: Principal Component Analysis (PCA)

Step-by-Step Logic:

1. **Collect the Data** – Collect customer data such as **Purchase Frequency, Total Spending, Average Order Value, Website Visits** and **Cart Abandonment Rate**.
2. **Prepare the Data** – Handle missing values and unnecessary data.
3. **Scale the Features** – Standardize the 200+ features.
4. **Apply PCA** – Use **PCA** to convert the large number of features into fewer principal components.
5. **Keep Important Information** – Select components that retain most of the important information.
6. **Train and Predict** – Train the model using the reduced features and predict customer behavior.

Scenario 4 – Collaborative Filtering

Question:

A digital library wants to recommend books to readers based on what similar readers liked. How should they design this system?

Concept: Collaborative Filtering

Step-by-Step Logic:

1. **Collect the Data** – Collect **Reader ID, Book Name, Rating** and **Reading History**.
2. **Create User-Item Matrix** – Represent readers as rows and books as columns, with ratings as values.
3. **Find Similar Readers** – Compare readers based on their ratings and identify readers with similar interests.
4. **Find Liked Books** – Identify books that similar readers rated highly.

5. **Recommend Books** – Recommend those books to the target reader that they have not already read.
6. **Predict and Recommend** – Rank the books based on predicted preference and recommend the best ones.

Scenario 5 – Feature Selection

Question:

A bank wants to assess the risk level of credit applicants using only the most important financial indicators. How can they reduce the number of features?

Concept: Feature Selection

Step-by-Step Logic:

1. **Collect the Data** – Collect credit applicant data such as **Income, Credit Score, Loan Amount, Age, Debt-to-Income Ratio** and **Credit Risk**.
2. **Separate X and y** – Use financial indicators as **Features (X)** and credit risk as the **Target (y)**.
3. **Apply Feature Selection** – Use **SelectKBest** with a suitable scoring method to identify the most relevant features.
4. **Select Important Features** – Rank the features and select only the top important indicators, such as **Credit Score, Income and Debt-to-Income Ratio**.
5. **Train the Model** – Train the credit risk model using only the selected features.
6. **Predict Risk** – Use the trained model to predict the risk level of new credit applicants.

Scenario 6 – Hybrid Recommendation System

Question:

A news app wants to recommend articles based on both article similarity and user reading history. How can they implement a hybrid system?

Concept: Hybrid Recommendation System

Step-by-Step Logic:

1. **Collect the Data** – Collect **Article Title, Category, Keywords** and **User Reading History**.

2. **Find Similar Articles** – Use article content to find articles similar to what the user has read.
3. **Analyze User History** – Check the user's previously read articles to understand their interests.
4. **Combine Results** – Combine **article similarity** and **user history** recommendations.
5. **Rank Articles** – Rank the articles based on the combined results.
6. **Recommend Articles** – Recommend the top relevant articles to the user.

Scenario 7 – Feature Selection

Question:

You're building a spam detection model and have thousands of text features from emails. How do you identify the most useful ones?

Concept: Feature Selection

Step-by-Step Logic:

1. **Collect the Data** – Collect email data such as **Email Text, Sender, Keywords** and **Spam/Not Spam**.
2. **Create Text Features** – Convert the email text into numerical features using **TF-IDF**.
3. **Apply Feature Selection** – Use **SelectKBest** to identify the most useful text features.
4. **Select Important Features** – Keep only the top features that are most relevant to spam detection.
5. **Train the Model** – Train the spam detection model using the selected features.
6. **Predict Spam** – Use the trained model to classify new emails as **Spam or Not Spam**.

Scenario 8 – Collaborative Filtering

Question:

An ed-tech platform wants to recommend courses based on what similar learners have enrolled in. What steps would you take?

Concept: Collaborative Filtering

Step-by-Step Logic:

1. **Collect the Data** – Collect **Learner ID, Course Name** and **Enrollment History**.

2. **Create User-Item Matrix** – Create a matrix with learners as rows and courses as columns.
3. **Find Similar Learners** – Compare learners based on their enrolled courses.
4. **Find Popular Courses** – Identify courses taken by learners with similar interests.
5. **Recommend Courses** – Recommend courses that similar learners have enrolled in.
6. **Predict and Recommend** – Rank the courses and recommend the most relevant ones to the target learner.

Scenario 9 – Dimensionality Reduction

Question:

You're developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine type), how do you reduce complexity?

Concept: Dimensionality Reduction

Step-by-Step Logic:

1. **Collect the Data** – Collect **Brand, Mileage, Engine Type, Year, Fuel Type** and **Price**.
2. **Prepare the Data** – Clean the data and handle missing values.
3. **Scale the Features** – Standardize the numerical features.
4. **Apply PCA** – Use **PCA** to reduce the 100+ features into fewer components.
5. **Keep Important Information** – Select components that retain most of the important information.
6. **Train and Predict** – Train the car price model using the reduced features and predict prices for new cars.

Scenario 10 – Cold Start Problem

Question:

How do you recommend products to new users who haven't interacted with anything yet?

Concept: Cold Start Problem

Step-by-Step Logic:

1. **Collect the Data** – Collect product data such as **Product Name, Category, Price** and **Rating**.

2. **Identify New Users** – Identify users who have no previous interaction or rating history.
3. **Use Popular Products** – Recommend products that are highly rated or frequently purchased.
4. **Use Basic Preferences** – If available, use the user's selected **Category or Interests** to improve recommendations.
5. **Rank Products** – Rank the products based on popularity and relevance.
6. **Recommend Products** – Recommend the top products to the new user and later use their interactions for personalized recommendations.